

Design review #4 – Space graded INA



Project developed by:	SLICE Semiconductor
Project funded by:	nlnet foundation
Document ID:	DR4: Space graded INA
Created by:	Diarmuid Collins
Date created:	28/08/2026
Date modified:	
Revision:	A

Table of Contents

1	General information	2
1.1	Project goal.....	2
1.2	Purpose of Design review #4 (DR4)	2
1.3	Repository details	2
1.4	Open source tools.....	3
1.5	Open source PDK.....	3
2	Pinlist	4
3	Package	5
4	Padring and sealring.....	6
4.1	Padring.....	6
4.2	Sealring	7
4.3	Padring and seal ring	9
5	Top level layout	10
6	Dummy fill generation	12
7	Final DRC sign off	14
8	Final top level simulations	15
9	Final tapeout submission	16
10	Post silicon lab characterization.....	17
11	Power up procedure	18
12	References	19
13	Revision History	19

1 General information

This project will develop an instrumentation amplifier (INA) with programmable gain (through I2C / SPI interfaces) which achieves high accuracy (< 1% gain error), low noise (< 50nV/rtHz at 1kHz), and high common mode rejection (> 80dB). It will produce an application specific integrated circuit (ASIC) using free and open tools which will be taped out on IHP's 130nm open source PDK, with performance guaranteed across process (slow / fast silicon), voltage ($\pm 5\%$) and temperature. Because of its high performance, the INA will be applicable to high accuracy sensing applications (pressure, strain, temperature) typical to those found in a wide variety of industry / medical applications. In addition, the INA will be made robust to high radiation environments, making it applicable to low earth orbit / high energy particle applications, where single even latch up can be problematic. The final chip will be characterised as a tangible proof of the maturity of today's free and open source toolchains and their readiness to produce high performance semiconductors.

1.1 Project goal

To show the capability of open source tools to develop industry quality ASICs, verified across industry standard conditions and taped out on silicon from a European fab (IHP).

To validate this goal, targeted performance will match that of an INA from a leading industry vendor (AD8223A [1]).

1.2 Purpose of Design review #4 (DR4)

Purpose of DR4 is to review the final ASIC before delivery to fab (i.e. tapeout) integration. Therefore, upon approval of DR4, ASIC is "frozen" and taped out.

1.3 Repository details

All material relating to this project can be found in the below publicly available repository:

https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main

Screenshot of the project landing page is shown in Figure 1.

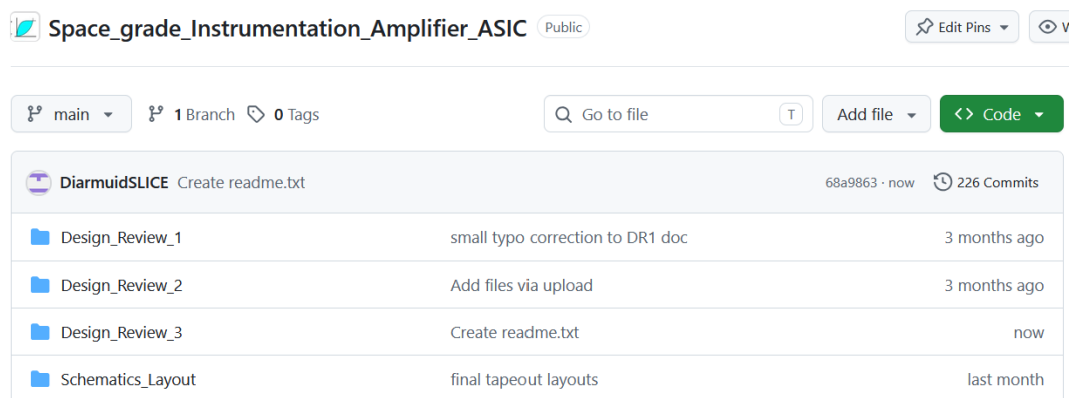


Figure 1 Project landing page

As this document pertains to DR4, all material relating to it is found in the sub directory "Design_Review_4". As block schematics relate to all DR's, these are contained in the separate folder "Schematics_Layout". As shown in Figure 2 this folder contains a sub-directory to each block in the ASIC design.

Note: Figure 2 shows only a small portion of the total number of blocks contained in this folder.

Inside each folder here, lives the following views:

- Schematic (frozen at DR2)
- Symbol (frozen at DR2)
- Layout (frozen at DR3)

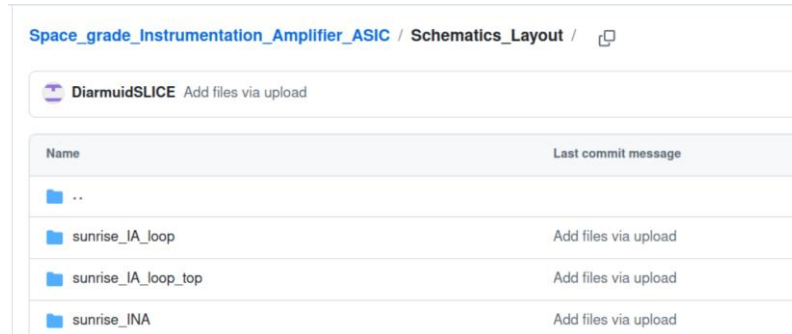


Figure 2 Project design blocks

1.4 Open source tools

The open source tools used for the ASIC are summarized in Table 2, with links to install each tool provided as references. In addition, tool versions used for the tapeout are included to enable results to replicated exactly at a later date.

Tool	Description	Version
ngspice	Spice simulator [2]	
xschem	Schematic / waveform viewer [3]	
openVAF	verilogA compiler [4]	
Octave	Post processing [5]	
klayout	Layout viewer / editor [6]	
magic	Pcell generator / PEX generation [7]	
netgen	LVS [8]	

Table 1: Open source tools used for this ASIC

1.5 Open source PDK

The open source process development kit (PDK) used for the design is from IHP, process option – SG13CMOS5L [9]. This is a derivative of their more general option – SG13G2 [10] as it excludes the following:

- Top metal layer 2
- Linear caps (MiM or MoM)
- Deep Nwell (DNW)
- HBT devices

DR4: Space graded INA

The first 3 items impacted the design but not the 4th since the design is not RF. SG13CMOS5L option was chosen primarily because its tapeout date of 13/07/26 suited the project timeline. In addition, it is lower cost (max = €1.5k/mm² as opposed to €2.8k/mm² for SG13G2) and runs though the fab quicker (5 months instead of 6) to achieve a fab out date in 16/12/26 [11].

2 Pinlist

Full pinlist of the ASIC was extensively reviewed in [12] resulting in the list summarized in Table 2.

#	Name	Description	IO Type	Domain	Value
1	SPI_CPOL	Mode select (<i>or rdac_ctrl[1]</i>)	IP	Digital	0 / vdda_hv
2	SPI_CPHA	Mode select (<i>or rdac_ctrl[0]</i>)	IP	Digital	0 / vdda_hv
3	I2C_SCL	I2C serial clock lin (<i>or cdac_sel[4]</i>)	IP	Digital	0 / vdda_hv
4	I2C_SDA_O	I2C data out	TRI	Digital	0 / vdda_hv
5	I2C_SDA_I	I2C data in (<i>or cdac_sel[3]</i>)	IP	Digital	0 / vdda_hv
6	SPI_nCS	SPI chip select (<i>or INA pd</i>)	IP	Digital	0 / vdda_hv
7	SPI_SCLK	SPI serial clock line (<i>or int bias</i>)	IP	Digital	0 / vdda_hv
8	SPI_MOSI	SPI MOSI data line (<i>or ip_ctrl[1]</i>)	IP	Digital	0 / vdda_hv
9	SPI_MISO	SPI MISO data line	TRI	Digital	0 / vdda_hv
10	SPI_MISO_CTRL	SPI MISO tri-stat ctrl bypass (<i>or ip_ctrl[0]</i>)	IP	Digital	0 / vdda_hv
11	CLK_EXT	External clock	IP	Digital	0 / vdda_hv
12	RO_OUT	RO output	OP	Digital	0 / vdda_hv
13	CLK_INT_EN	Pass RO o/p to digital	IP	Digital	0 / vdda_hv
14	RO_EN	Enable the RO	IP	Digital	0 / vdda_hv
15	VTSNS2_N	Temp sensor bias	ANA	Analog	1 / 10uA
16	VTSNS2_P	Temp sensor bias	ANA	Analog	1 / 10uA
17	VREF	Voltage VOUT is referred to	ANA	Analog	0 – Vref_max
18	VOUT	INA output	ANA	Analog	50mV – 3.3-50mV
19	VSSD	Digital ground	Supply	Digital	0V
20	VDDA_LV	Low voltage supply	PWR	Digital	1.2 – 1.5V
21	IBIAS_IN_5U	External bias	ANA	Analog	5uA
22	INT_BIAS	Sense int_bias internal signal	OP	Digital	0 / vdda_hv
23	PORB	Power on reset (active lo)	IP	Digital	0 / vdda_hv
24	SERIAL_SEL	Select between SPI and I2C (<i>or cdac_sel[5]</i>)	IP	Digital	0 / vdda_hv
25	TEST_MODE	Repurpose SPI / I2C controls as direct i/p's	IP	Digital	0 / vdda_hv
26	VSSA	Analog ground	GND	Analog	0V
27	VDDA_HV	High voltage supply	PWR	Analog	3.3V
28	VSNS_EXT_N1	External sense (-ve)	ANA	Analog	0.9 – Vcm_max_n

#	Name	Description	IO Type	Domain	Value
29	VSNS_EXT_P1	External sense (-ve)	ANA	Analog	0.9 – Vcm_max_n
30	VTSNS1_N	Temp sensor bias	ANA	Analog	1 / 10uA
31	VTSNS1_P	Temp sensor bias	ANA	Analog	1 / 10uA
32	VSNS_EXT_N2	External sense (-ve) with 260Ω CDM protection	ANA	Analog	0.9 – Vcm_max_n
33	VSNS_EXT_P2	External sense (+ve) with 260Ω CDM protection	ANA	Analog	0.9 – Vcm_max_p
34	I2C_ADDR[0]	Address (or cdac_sel[0])	IP	Digital	0 / vdda_hv
35	I2C_ADDR[1]	Address (or cdac_sel[1])	IP	Digital	0 / vdda_hv
36	I2C_ADDR[2]	Address (or cdac_sel[2])	IP	Digital	0 / vdda_hv

3 Package

Table 2 Top level pinlist

As shown in Table 2 a total of 36 pins are required. As a result, final ASIC will be delivered in a QFN36 package, provided by Serma [13] whose pin locations are shown in Figure 3.

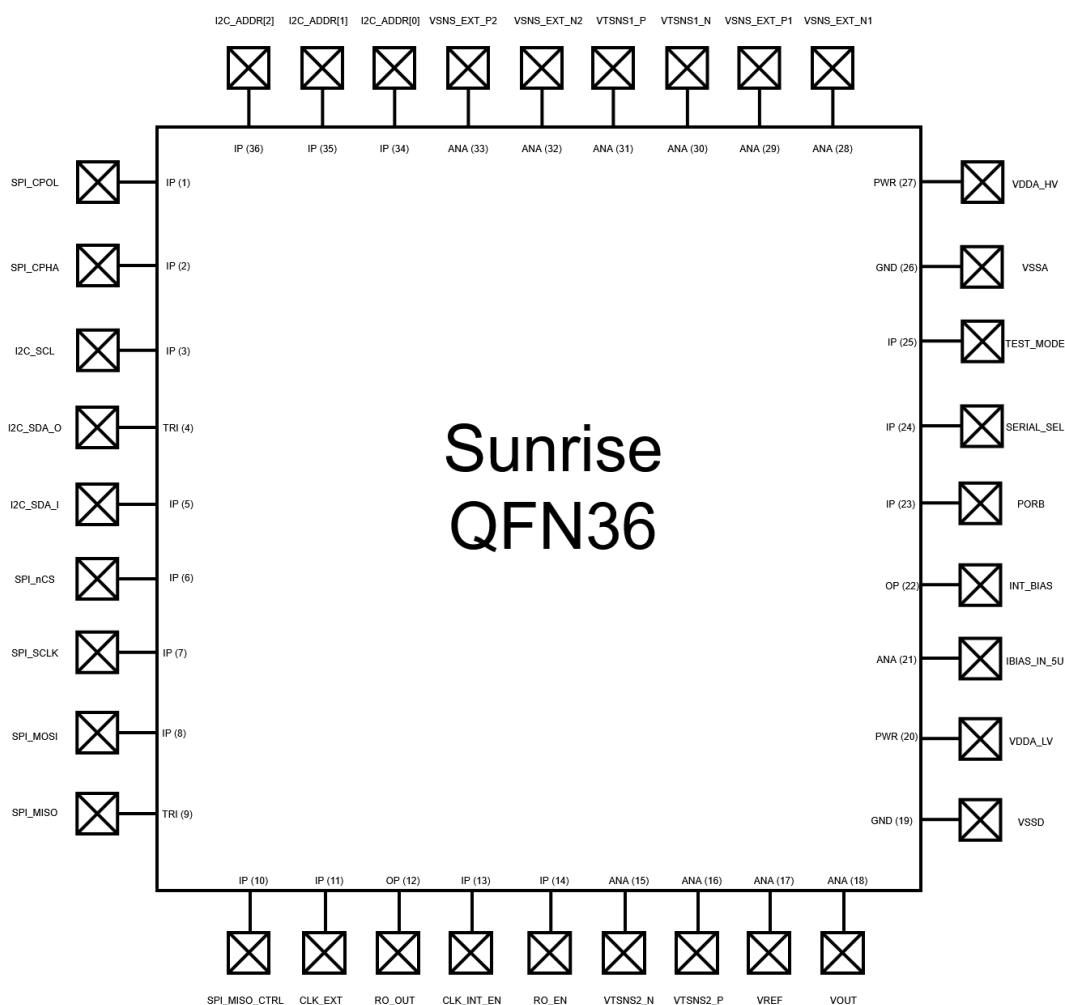


Figure 3 Sunrise pin placement

It should be noted that this pin placement drove the ASIC top level floorplan as shown in section 5.

4 Pading and seairing

4.1 Pading

All pins in Figure 3 must connect to internal IO cells which provide suitable ESD protection (typically 4kV HBM). As can be seen in Figure 3, a total of 8 types of IC cell are used, as summarised in Table 3 *Table* .

#	Cell name	Description
1	sunrise_pad_analog_io	analog io
2	sunrise_pad_digital_i	digital input
3	sunrise_pad_digital_o	digital output
4	sunrise_pad_digital_tri	digital tri state
5	sunrise_pad_pwr_iovdd	io supply (3v3)
6	sunrise_pad_pwr_iovss	io gnd
7	sunrise_pad_pwr_vdd	core supply (1v5)
8	sunrise_pad_pwr_vss	core gnd

Table 3 iocells used in sunrise

All the above cells are provided by IHP and can be downloaded from [14]. Documentation on the cells can then be found at [15]. To give a better understanding of how these cells are implemented in sunrise, the analog_io and digital_i cells are shown in Figure 4.

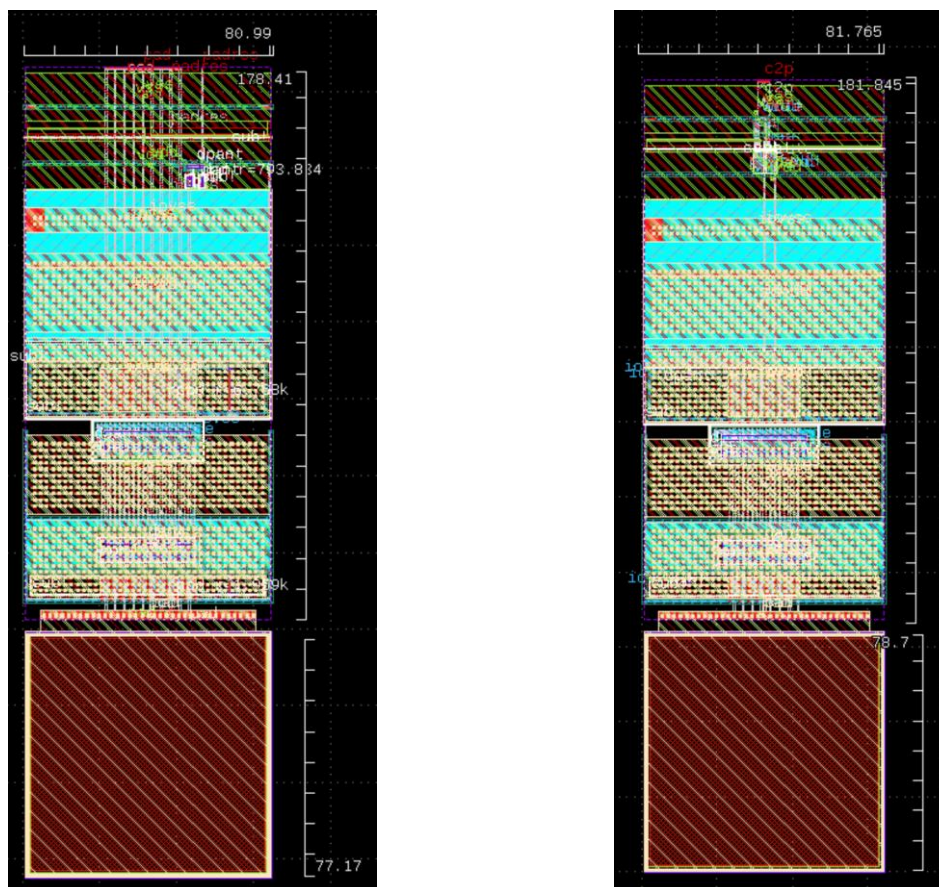


Figure 4 a) Analog io cell + bondpad

b) Digital input cell + bondpad

DR4: Space graded INA

In Figure 4, the IO cells are shown to occupy a footprint of $\sim 80 \times 180 \mu\text{m}$. Also shown is the $80 \times 80 \mu\text{m}$ square bondpad connected to the IO cell which connects to an external pin.

Each of the 8 cells have been extracted and individually verified to be working as expected. For digital cells this meant toggling inputs and observing outputs, or vice versa. For analog cells it meant placing those cells on the relevant pins in the testbench to verify analog performance was not impacted.

Actual ESD robustness of the cells was verified in [16] for the S8 process, which also applies to 5L since the cells only differ by the top metal layer.

Finally the IO cells were connected to form a padding compatible with the pinout of Figure 3. Resulting padding is shown in Figure 5.

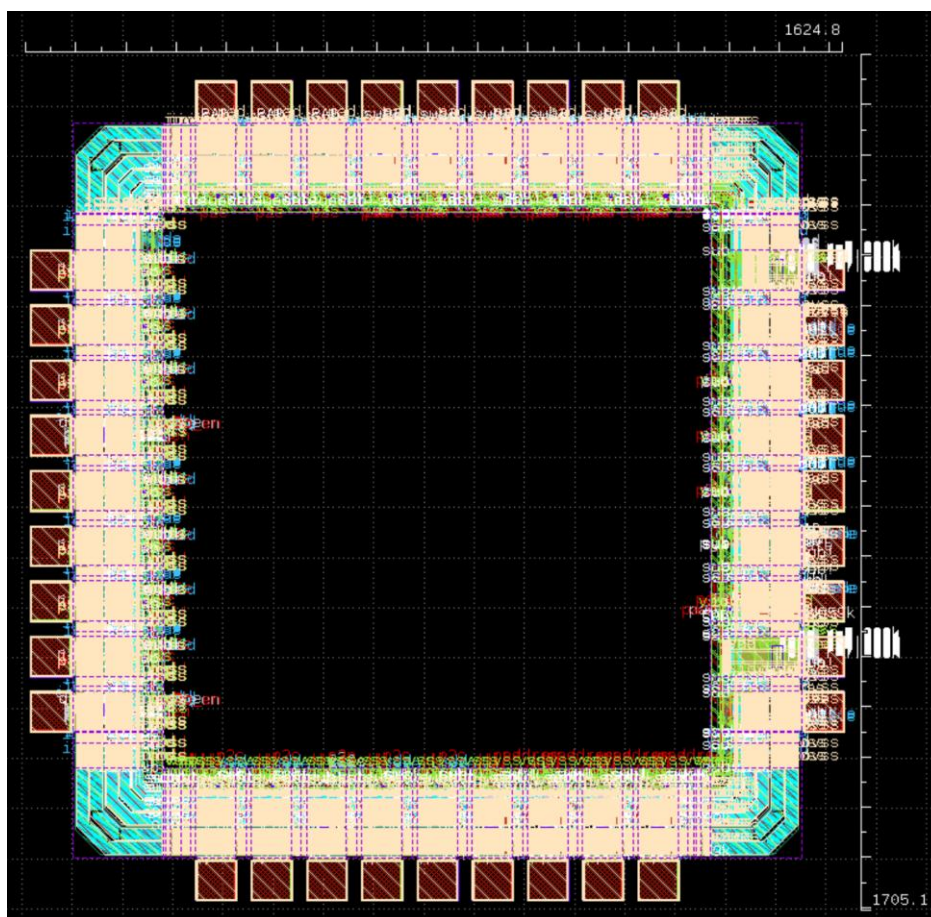


Figure 5 sunrise padding

It should be noted that when creating the padding, only the layouts of the IO cells were available, not their corresponding schematics. As a result, LVS could not be performed on the padding.

Since taping out, we have been informed by the JKU that such schematics are available at [17]. Therefore, post tapeout we plan to download these spice netlists, attach them to a symbolic view of the padding and perform LVS on that. There are various steps involved here so it will take some time however.

Not having LVS'd the padding is seen as low risk since it was closely inspected visually, multiple times, using the probe tool in klayout.

4.2 Seairing

The seairing defines the physical boundary of the die and thus defines the final dimensions of it. Seairing for sunrise is shown in Figure 6. Final dimensions taken from this seairing are shown in Figure 7 to be $\sim 1.75 \times 1.75 \text{ mm}$ to give a total area of $\sim 3.1 \text{ mm}^2$.

Valid from:	Document ID:	Rev.	Page:	
28.08.2026	DR4: Space graded INA	A	7/19	

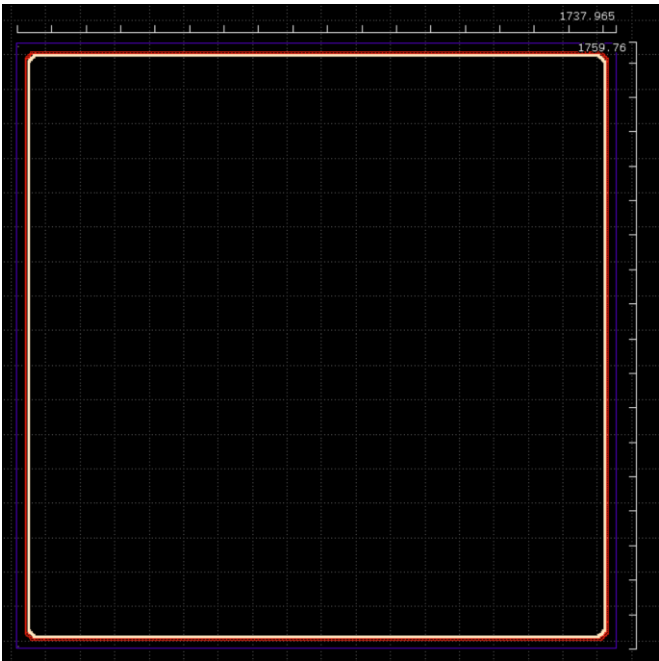


Figure 6 sunrise sealing

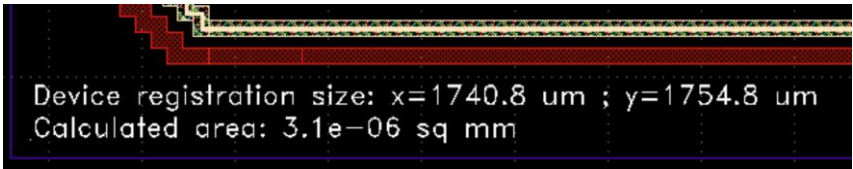


Figure 7 sunrise dimensions

With the IHP sealing it is important to make the distinction between the EdgeSeal.boundary and EdgeSeal.drawing layers.

The EdgeSeal.boundary layer defines the boundary of the die where it will be diced from the wafer. The EdgeSeal.drawing layer defines the boundary of the actual circuitry on the die. Metal fill and corresponding density calculations must go to the outer edge of the EdgeSeal.drawing layer while, due to the cutting nature of the dicing process, the region between the EdgeSeal.drawing and EdgeSeal.boundary layers must be left empty.

Figure 8 shows the two layers which unfortunately are given the same colour making them a little difficult to distinguish. Therefore, Figure 9 shows a close up of the two layers to better show their difference.

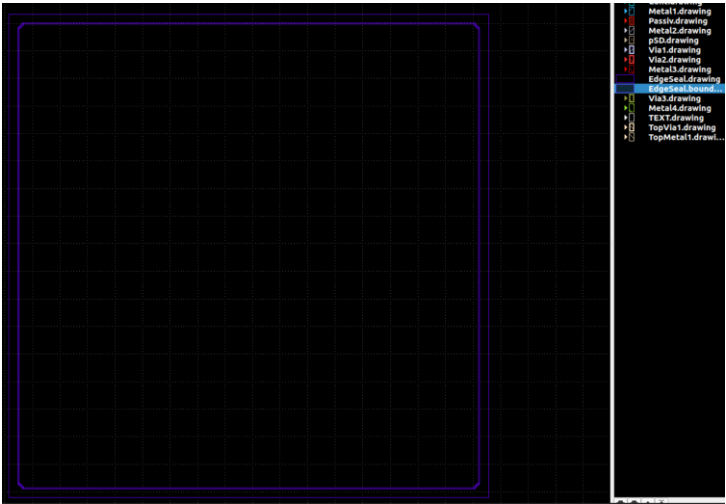


Figure 8 EdgeSeal.drawing and EdgeSeal.boundary layers

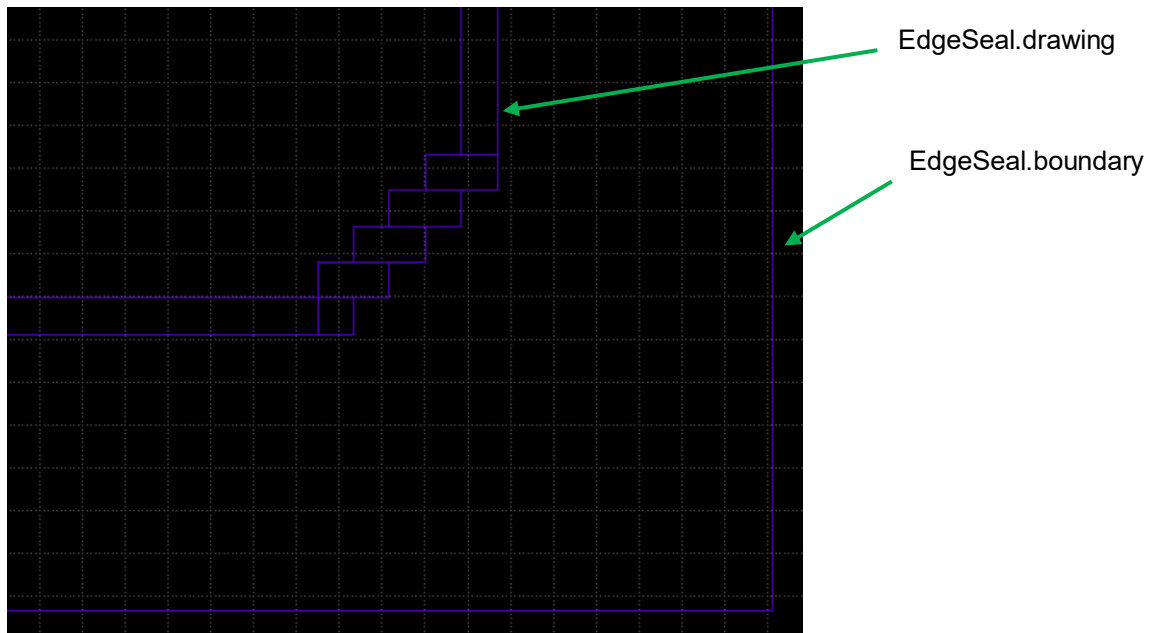


Figure 9 Close up of *EdgeSeal.drawing* and *EdgeSeal.boundary* layers

4.3 Pading and seal ring

For completion, Figure 10 shows the padding and sealing of sunrise combined.

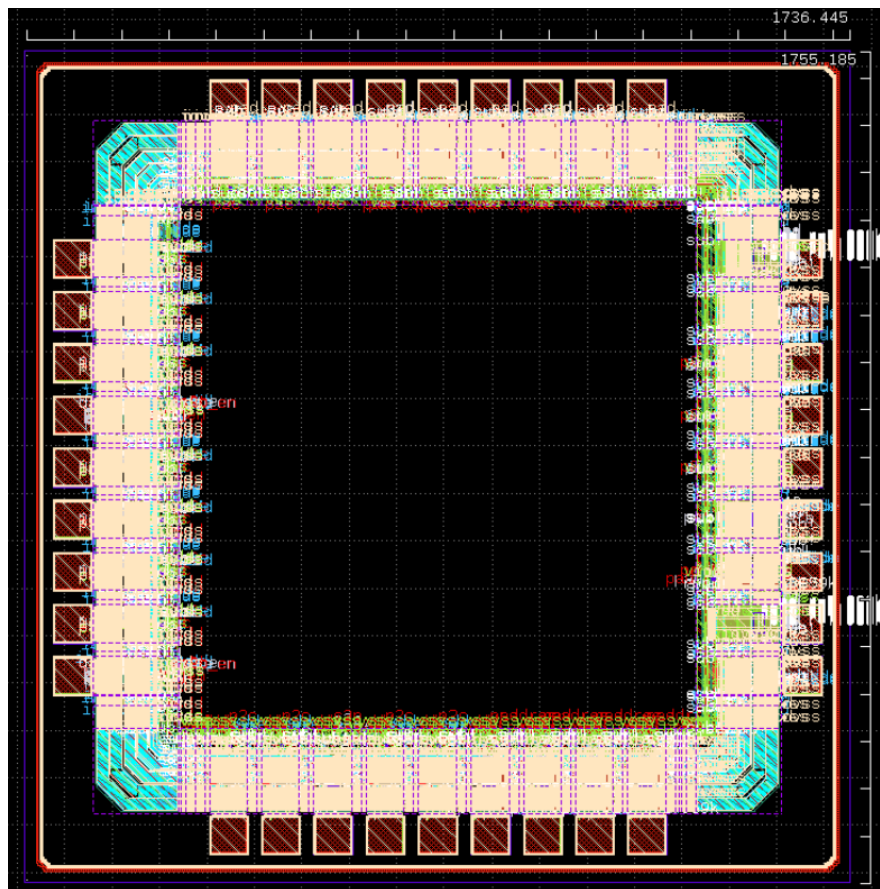


Figure 10 sunrise padding and sealing combined

5 Top level layout

Top level layout is shown in Figure 11 where it can be seen to occupy a footprint of $\sim 1 \times 1 \text{ mm}$. Comparing with the pin placement of Figure 3 one can see how that pin placement drove the top level floorplan (e.g ring oscillator sits in the bottom left corner close to all its control pins, main temp sensor sits in the top center close to its inputs, digital sits towards the left hand side close to its control pins, etc).

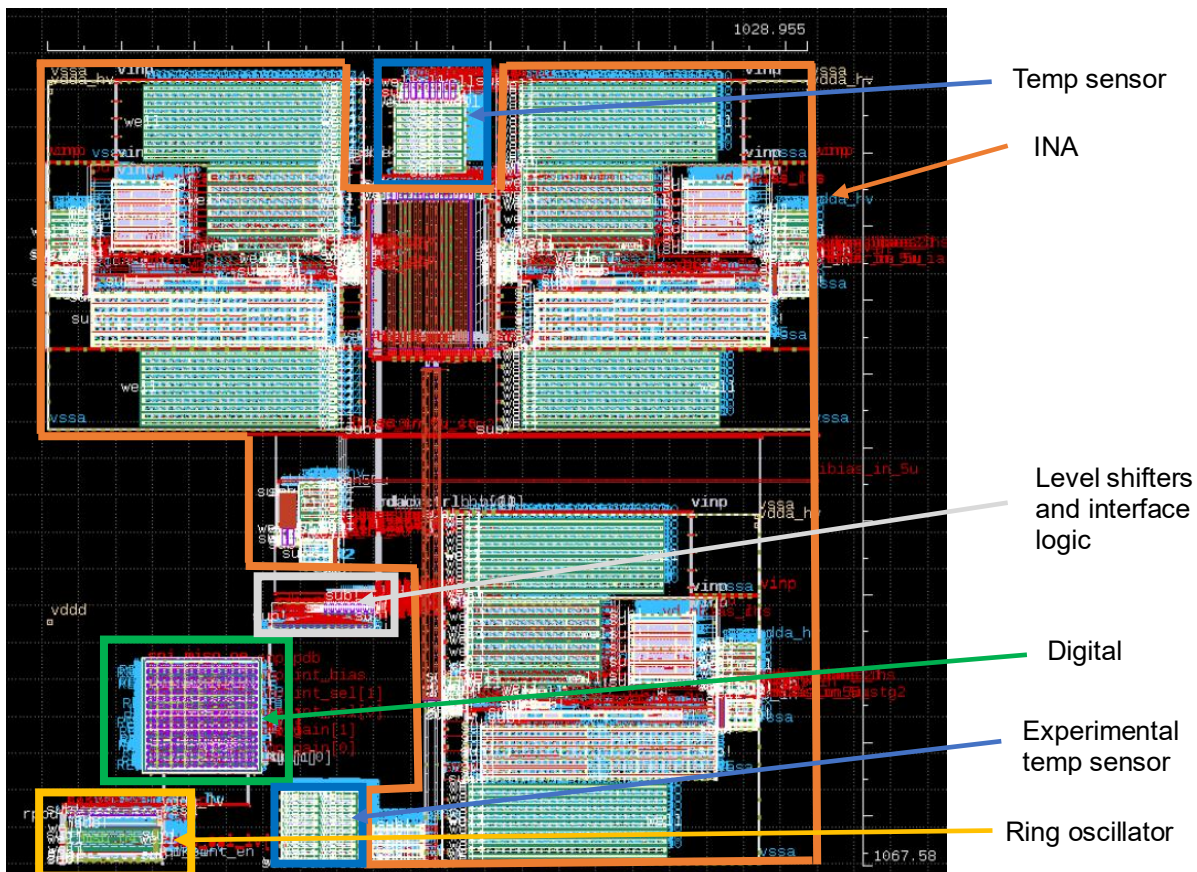


Figure 11 sunrise top level layout

A schematic view instantiating all the above blocks, apart from the digital was created and LVS'd against the corresponding layout. Performing LVS on a view including the digital is a difficult task, even in commercial tools. An approach has been proposed which we plan to investigate post layout.

Not having LVS'd the top level including the digital is seen as low risk since it was closely inspected visually, multiple times, using the probe tool in klayout.

It is hoped that the digital and padding post tapeout LVS activities will be successful, as the resulting flow will enable full chips for future projects to be LVS'd in their entirety, thus providing a very thorough signoff.

Top level layout including the power grid of M4/M5 is shown in Figure 12.

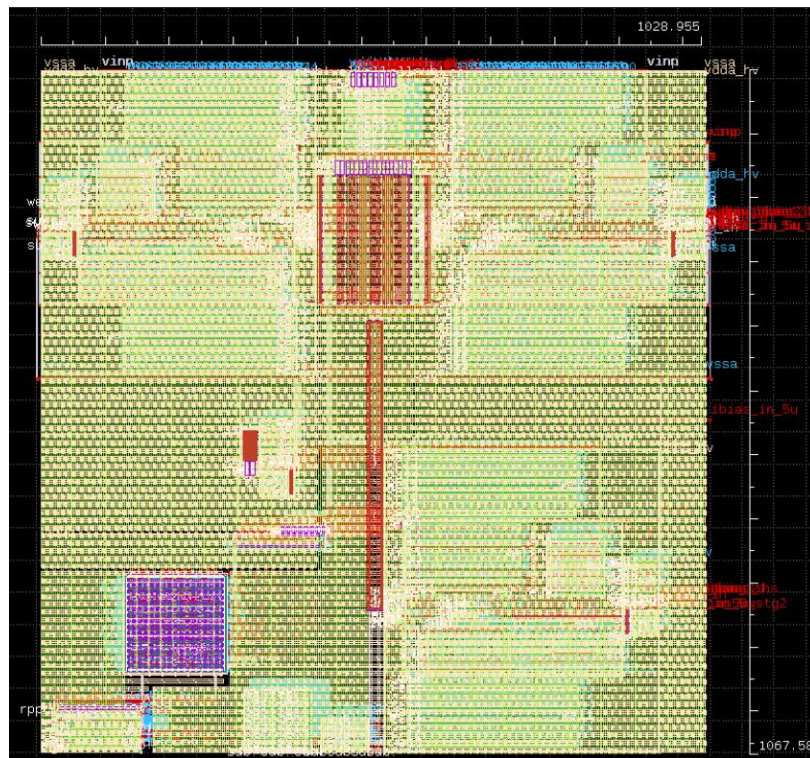


Figure 12 sunrise top level layout including the power grid

Finally, top level layout including the power grid, padding and seal ring is shown in Figure 13.

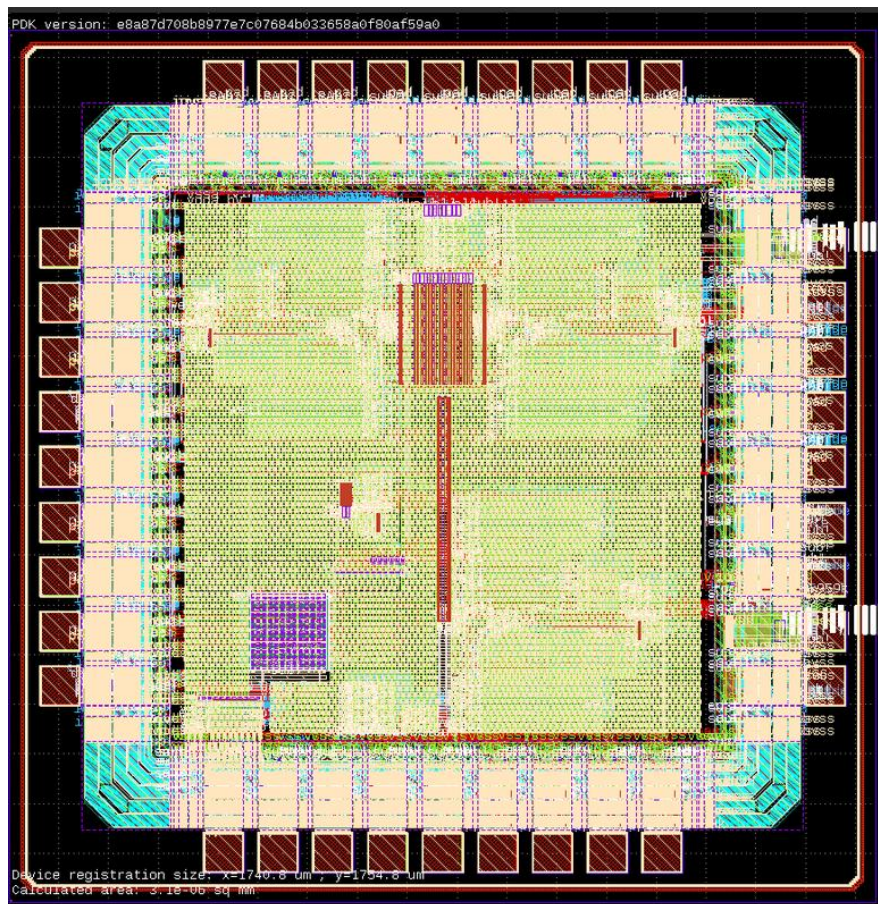


Figure 13 sunrise top level layout with power grid, padding and sealing

6 Dummy fill generation

To avoid over / under etching during the CMP steps of the fabrication process, all metal / poly / active densities must lie within a range specified by the foundry. To achieve this, dummy fill is generated and placed between all specified areas. Fill generation script is provided by the foundry which must not overfill creating max density violations, whilst at the same time place enough fill to avoid min density violations.

As dummy fill can affect matching, exclusion zones are placed above matching sensitive analog circuitry where the fill algorithm knows not to place any fill. An example of this is shown in Figure 14 where M1 fill has been applied in all required spaces except above the matching critical analog circuit which has been defined by the M1 fill block (Metal.nofill).

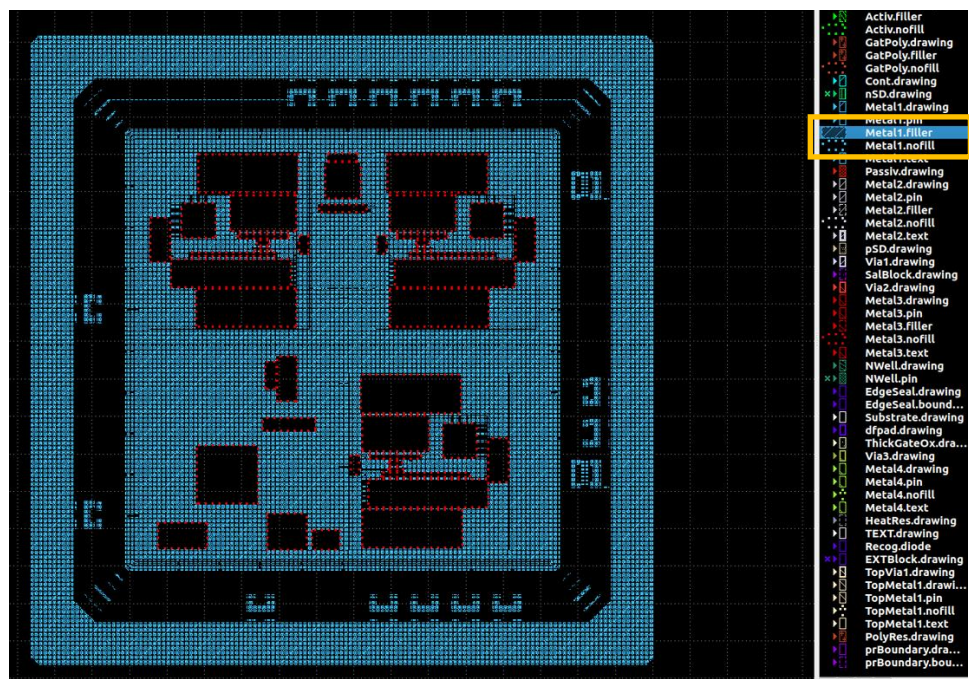


Figure 14 M1 fill

Metal fill spacings were specified as per Figure 15.

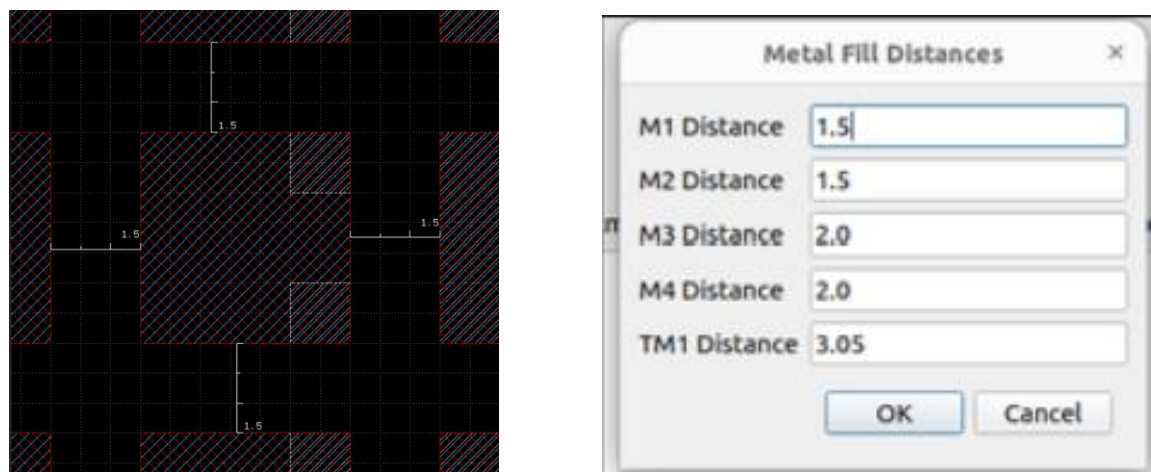


Figure 15 Metal fill spacings

Resulting fill of all the layers at the top level is shown in Figure 16, which represents the final top level view of the sunrise ASIC.

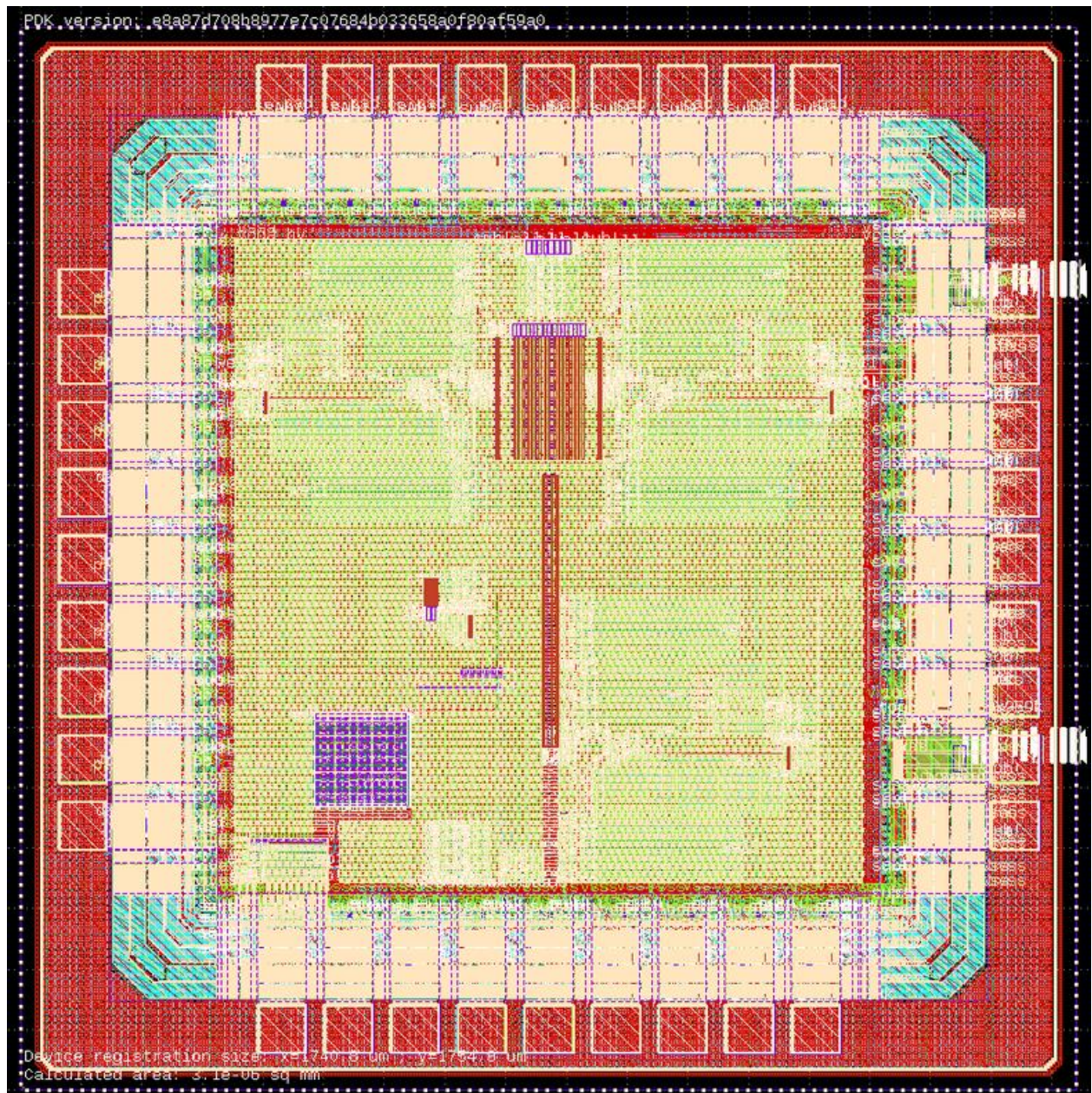


Figure 16 sunrise final ASIC

Note in Figure 16 how no fill is applied between the EdgeSeal.drawing and EdgeSeal.boundary layers since this region is left blank because of the die cutting process as described in section 4.

Prior to tape out, DRC density check must be run on the top level layout. DRC density checker from IHP was found to have the below bugs:

- Included up to the EdgeSeal.boundary and not the EdgeSeal.drawing
- Double counted (e.g. treated 2 identical metals placed on top of each other as 2 separate metals)

These bugs resulted in incorrect density results. In particular, the 2nd bug resulted in density numbers $\gg 100\%$ as shown in Figure 17.


```

Layer areas and percentages (units: um^2)
-----
Total Area      = 3856.31 um^2

Activ   Area      = 1476664.54 um^2
Activ   Density % = 38292.14% [Min = 35.00%, Max =

GatPoly Area      = 577226.20 um^2
GatPoly Density % = 14968.34% [Min = 15.00%]

Metal1  Area      = 1418750.71 um^2
Metal1  Density % = 36790.34% [Min = 35.00%, Max =

Metal2  Area      = 1126023.13 um^2
Metal2  Density % = 29199.48% [Min = 35.00%, Max =

```

Figure 17 Incorrect density calculations from the IHP script

To address this, Yacine Sotehi investigated and fixed these bugs. Re-running the top level layout using this updated script showed much more reasonable results, provided in Figure 18.

```

#####
:          Density Report          #
: #####
:
: Layer areas and percentages (units: um^2)
: -----
:
: Total Area      = 2833080.96 um^2
:
: Activ   Area      = 1476664.54 um^2
: Activ   Density % = 52.12% [Min = 35.00%, Max = 55.00%]
:
: GatPoly Area      = 577226.20 um^2
: GatPoly Density % = 20.37% [Min = 15.00%]
:
: Metal1  Area      = 1418750.71 um^2
: Metal1  Density % = 50.08% [Min = 35.00%, Max = 60.00%]
:
: Metal2  Area      = 1125930.58 um^2
: Metal2  Density % = 39.74% [Min = 35.00%, Max = 60.00%]
:
: Metal3  Area      = 1444605.59 um^2
: Metal3  Density % = 50.99% [Min = 35.00%, Max = 60.00%]
:
: Metal4  Area      = 1326832.28 um^2
: Metal4  Density % = 46.83% [Min = 35.00%, Max = 60.00%]
:
: TopMetal1 Area      = 1396418.82 um^2
: TopMetal1 Density % = 49.29% [Min = 25.00%, Max = 70.00%]
:
:

```

Figure 18 Correct density calculations from the updated IHP script

These results were verified as they matched those provided by calibre (the gold standard for pre-tapeout checks).

To use this updated, the user must do the following:

- Go to your \$PDK_ROOT area
- git clone <https://github.com/YacineSot/ihp-sg13cmos5l> <some_name>
- Set \$PDK to <some_name>
- Run DRC density check as normal

7 Final DRC sign off

DRC of the entire top level ASIC had to be performed in magic since it kept crashing klayout. Suspected reason is the DRC deck rules are coded much more efficiently in magic thus requiring less memory to execute. Resulting top level DRC results are shown in Figure 19.

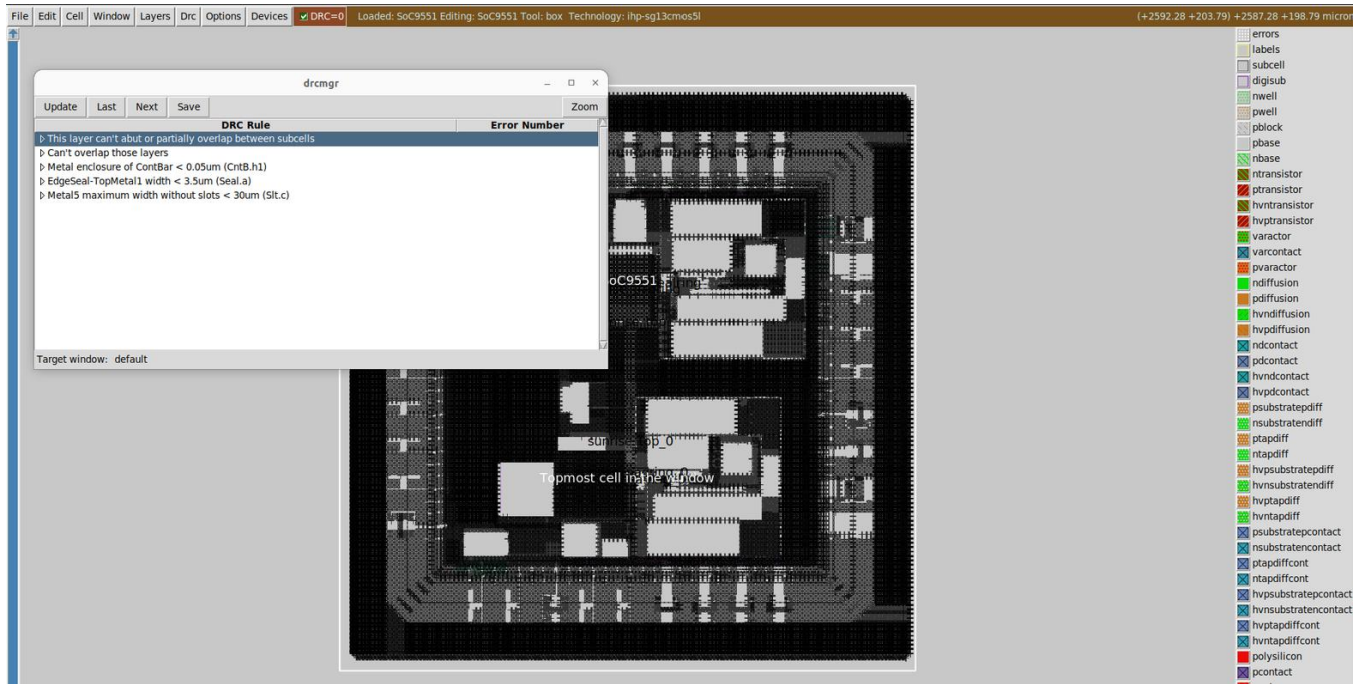


Figure 19 sunrise DRC signoff results from magic

A total of 5 DRC categories were reported. All were investigated and confirmed by IHP / magic tool owner (Tim Edwards) to be either waivable or due to a bug in the DRC checker. Therefore, the above constitutes a clean DRC result. Calibre (industry golden reference sign off for DRC) was next applied by IHP to the top level pre-tape out, which verified this clean result.

8 Final top level simulations

Although all blocks were simulated extensively post layout in their respective DR3's, as a final sanity check, top level specs were simulated at TT/27C on an extracted netlist containing all analog blocks. Results are summarized in Table 4.

	SCH	RCX
CMRR (dB)	-114.372	-114.784
GE (%)	-0.117	-0.117
fosc (MHz)	10.45	9.82
D_osc (%)	52.48	51.02
idda (uA)	785.7	785.6
BW (kHz)	301.4	300.1
Terror (degC)	0.697	0.699
vout_sample (V)	1.7	1.7

Table 4 Final top level simulation results pre (SCH) and post (RCX) layout

As shown in Table 4, pre and post layout results match very well to completely verify top level performance and hook up.

DR4: Space graded INA

It should be noted that the digital block was left out of this analysis. This is because the co-sim environment to simultaneously test spice and Verilog netlists was not set up pre-tapeout. Although an involved task, this will be investigated post tapeout since it represents a gap in the verification. This gap is deemed low risk since the Verilog has been thoroughly verified pre and post silicon with its hook up to the analog being closely inspected visually, multiple times, using the probe tool in layout.

9 Final tapeout submission

Final tapeout was committed to the below repo [18] as requested by IHP.

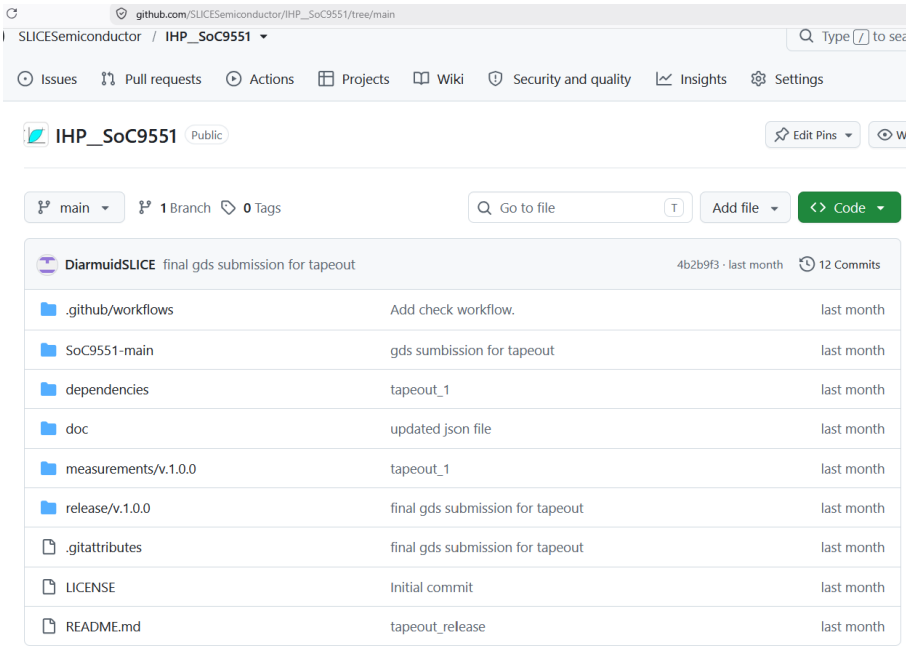


Figure 20 Sunrise tapeout repo at IHP

Tapeout requested was then submitted at [19] where all tapeout checks were completed / signed off on 16.07.26 as shown below.

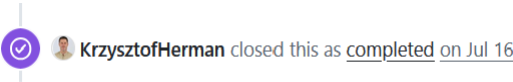


Figure 21 Final tapeout confirmation from IHP

10 Post silicon lab characterization

Table 5 summarises the equipment required to characterize this part post silicon. It should be noted that at the time of writing, all required equipment apart from the oscilloscope, have been acquired. We are currently investigating the most suitable oscilloscope for the best price.

Equipment		
Abbreviation	Description	Model
PSU	Power supply unit	Agilent E3631A
SMU	Source meter unit	Keithley 2602B
DMM	Digital multimeter	Agilent 34970A
SIGGEN	Signal generator	Keysight 33500B
SPEC	Spectrum analyser	Anritsu MS2691A
OSC	Oscilloscope	tbd
TC	Thermocouple	K-type

Table 5 Lab char equipment list

Building on Table 5, Table 6 summarises all the specs which will be characterized post silicon. It provides a brief description of how each spec will be measured, equipment used to make that measurement and char board requirements to facilitate the measurement.

Spec	PSU	SMU	DMM	SIGGEN	SPEC	OSC	TC	Char board requirements	Brief measurement description
GE	Y	Y	Y	Y	N	N	N	Sense pts at INA i/ps	Measure output for various discrete diff i/ps + also a ramped i/p
CMRR	Y	Y	Y	N	N	N	N	Sense pts at INA i/ps	Measure output for a given diff i/p at different common mode levels
Noise	Y	Y	Y	N	Y	N	N	Low noise BNC at INA o/p	Measure noise density with INA gain set to 21
CSA	Y	Y	Y	N	N	N	N	Resistor dividers (x2) which can be connected to INA i/ps using jumpers	Keithley sources current through resistor divider ($r_{top} = r_{sense}$) which sends a diff voltage into the INA at a specified common mode level. Measure o/p and correlate result with sense resistor.
fosc	Y	N	N	N	Y	Y	N	BNC at RO o/p	Measure fosc using oscilloscope and spectrum analyser
fosc res	Y	N	N	N	Y	Y	N	BNC at RO o/p	Measure fosc for trim code increments
trim range	Y	N	N	N	Y	Y	N	BNC at RO o/p	Measure fosc at min and max trim codes
duty cycle	Y	N	N	N	N	Y	N	BNC at RO o/p	Measure duty cycle on the OSC
Tsns	Y	Y	Y	N	N	N	Y	Sense pts at Tsns i/ps Automated switching of i/p's before sense pts	Measure dvgs and using the simulated $T(vgs)$ transfer function, back calculate the junction temperature (T_j). Correlate with thermocouple ambient measurement (T_a), accounting for self-heating ($T_j = T_a - P.R_{th}$)
Power	Y	Y	Y	N	N	Y	Y	Sense pts at supply inputs	Power up ASIC and read current from PSU

Table 6 Post silicon char plan

Color coding scheme for Table 6 is explained in Table 7.

Legend
INA related measurement
RO related measurement
TS related measurement
Full ASIC measurement

Table 7 Legend for Table 6

It should be noted that the above focuses only on characterizing the analog performance of the ASIC. As the ASIC is programmable, the char board needs to include a microcontroller which can interface with the I2C/SPI. In addition, dip switches must also be included to enable the user to manually code the INA as a risk mitigation in the event of the I2C/SPI not functioning as expected. Finally, to facilitate bringing an external clock on board, a BNC connector must be included and connected to pin 11 in Figure 3 (CLK_EXT).

11 Power up procedure

The full digital to analog (D/A) interfacing of the ASIC is detailed in [20] which includes instructions on the correct sequence to power up the part. As this relates to lab char it is summarized here.

As detailed in [21], the level shifter was imported as IP which does not contain a reset. This introduces a risk whereby if vdda_lv comes up before vdda_hv, the level shifter could enter an undefined state leading to unwanted paths. To mitigate this it is vital the correct power up sequence is employed to guarantee vdda_lv comes up before vdda_hv. This is shown in Figure 22 and summarized in Table 8.

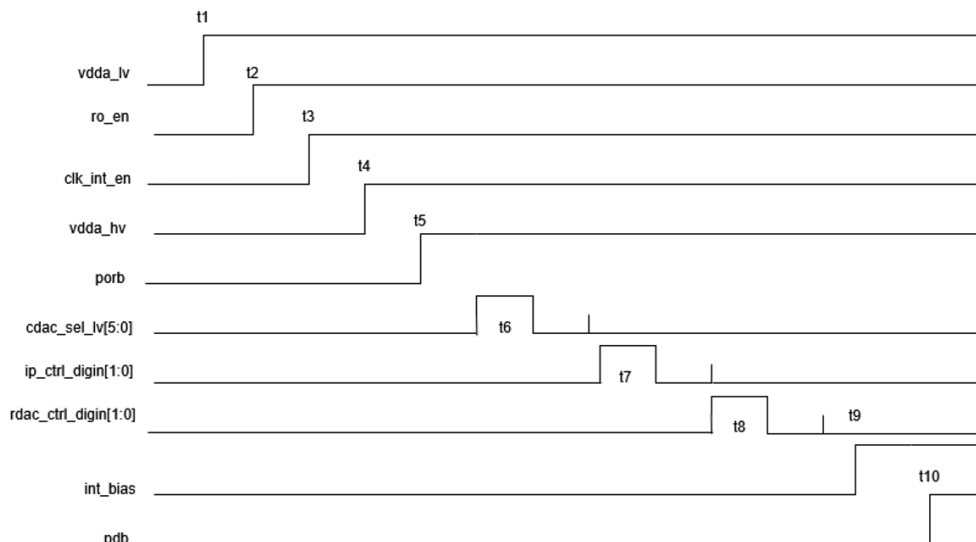


Figure 22 Power up sequence waveforms

Time point	What happens	Description	Comment
t1	vdda_lv ramps	Bring up the digital supply	
t2	ro_en asserts	RO starts oscillating	
t3	clk_int_en asserts	RO passes to the output / digital	If RO is not working, apply an external clk and set clk_int_en = 0
t4	vdda_hv ramps	Bring up the analog supply	Vdda_hv must come up after vdda_lv as level shifters have no reset
t5	porb asserts	Digital comes out of power on reset	Up to this point all digital o/p's = 0
t6	Program cdac_sel_lv[5:0]	Tune the RO 10MHz	Programmed through SPI / I2C
t7	Program ip_ctrl_digin[1:0]	Select the input to be sensed by the INA	Programmed through SPI / I2C
t8	Program rdac_ctrl_digin[1:0]	Set the gain of the INA	Programmed through SPI / I2C
t9	Set int bias	If using internal bias, int_bias = 1, if using external bias, int_bias = 0	External current source must only be turned on after int_bias = 0, or not at all
t10	Assert pdb	Take the analog out of reset	Since bias has been applied and all config bits set, analog can now be used

Table 8 Power up sequence summary

12 References

- [1] <https://www.analog.com/media/en/technical-documentation/data-sheets/AD8223.pdf>
- [2] <https://ngspice.sourceforge.io/download.html>
- [3] https://xschem.sourceforge.io/stefan/xschem_man/install_xschem.html
- [4] <https://openvaf.semimod.de/>
- [5] <https://octave.org/download>
- [6] <https://www.klayout.de/build.html>
- [7] <https://opencircuitdesign.com/magic/download.html>
- [8] <https://opencircuitdesign.com/netgen/>
- [9] <https://github.com/IHP-GmbH/ihp-sg13cmos5l>
- [10] <https://github.com/IHP-GmbH/IHP-Open-PDK>
- [11] <https://www.ihp-microelectronics.com/services/research-and-prototyping-service/mpw-prototyping-service/schedule-price-list>
- [12] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/blob/main/Design_Review_4/SUNRISE_Pinlist_Padring.docx
- [13] <https://www.serma-microelectronics.com/en/>
- [14] https://github.com/IHP-GmbH/ihp-sg13cmos5l/tree/main/libs.ref/sg13cmos5l_io/gds
- [15] https://github.com/IHP-GmbH/IHP-Open-PDK/tree/dev/ihp-sg13g2/libs.ref/sg13g2_io/doc
- [16] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/blob/main/Design_Review_4/Sunrise_padring.pdf
- [17] https://github.com/iic-jku/TinyWhisper/blob/main/ihp130/verification/lvs/sg13g2_io.spi
- [18] https://github.com/SLICESemiconductor/IHP_SoC9551/tree/main
- [19] <https://github.com/IHP-GmbH/Open-Silicon-MPW/issues/41#issuecomment-4955555848>
- [20] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/blob/main/Design_Review_4/SUNRISE_DA_INTERFACE_RevB.docx
- [21] https://github.com/SLICESemiconductor/Space_grade_Instrumentation_Amplifier_ASIC/tree/main/Design_Review_3/LVL_SHFT

13 Revision History

Rev.	Date	Author	Changes, Remarks
A	08.08.2026	Diarmuid Collins	Created

Table 9: Revision history